Butterfly Network

Dr.P.Ganesh Kumar
Anna University

Introduction

As parallel computers grew in size and complexity, traditional interconnection networks such as **bus**, **ring**, **mesh**, and **binary tree** became insufficient for supporting high-speed communication among thousands of processors. These topologies often suffer from long communication delays, bottlenecks, and poor scalability.

The **Butterfly Network** was developed as a **multistage interconnection network (MIN)** that provides efficient communication between processors using multiple switching stages. It enables data to travel from any source to any destination through a unique path while requiring significantly fewer communication links than a fully connected network.

Butterfly networks are widely used in **parallel computers**, **multiprocessor systems**, **network switches**, **packet routing**, **Fast Fourier Transform (FFT)** computations, and **high-performance communication systems** because of their low communication latency and logarithmic routing complexity.

Definition

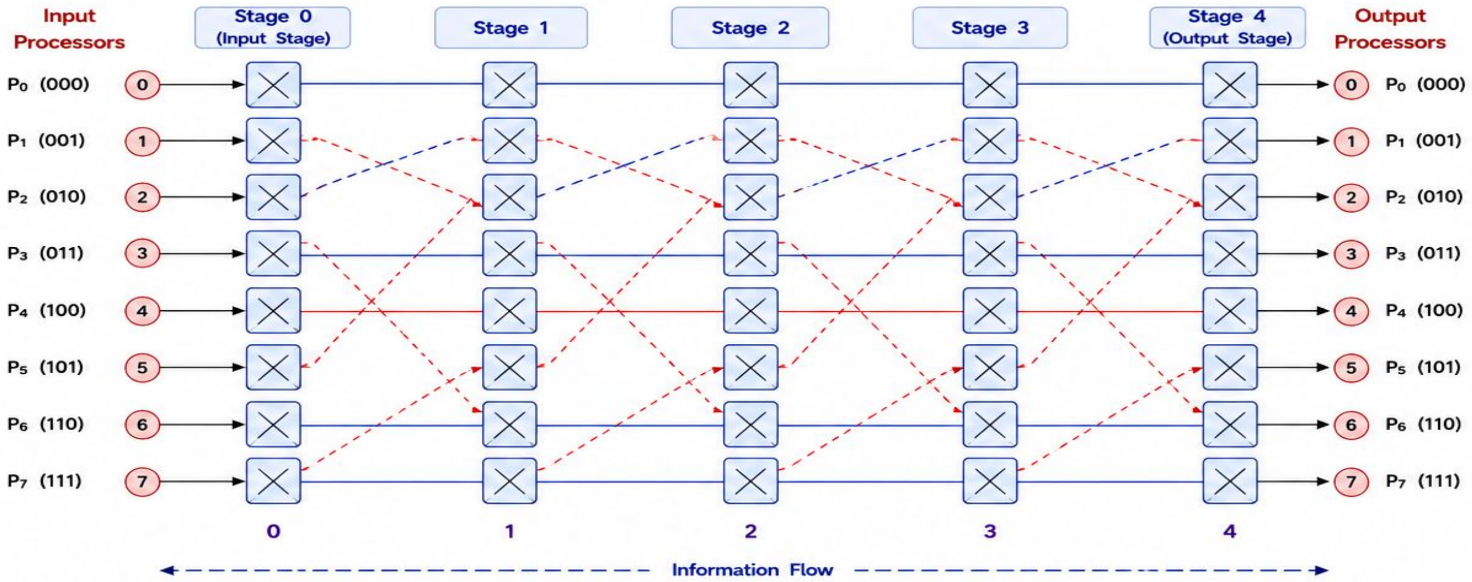
A **Butterfly Network** is a multistage interconnection network consisting of several stages of switching elements that connect processors through unique communication paths. Each stage performs controlled data exchanges, enabling efficient routing from any source node to any destination node.

Characteristics

- Multistage interconnection network (MIN)
- Logarithmic communication complexity
- Unique routing path
- Recursive architecture
- Highly scalable
- Suitable for parallel communication
- Efficient hardware utilization

Figure 6.1 – Basic Structure of Butterfly Network

A butterfly network is a multistage interconnection network consisting of switching elements arranged in $\log_2 N + 1$ stages to connect $N = 2^n$ input processors to N output processors.



Legend

- Processor (Input / Output)
- Switching Element (2 × 2)
- Straight Link (No change in bit)
- Cross Link (Bit change)

Example Shown:

$N = 8$ (2^3 Processors)

Number of Stages = $\log_2 8 + 1 = 4$ (Stages 0 to 4)

Each stage has $N/2$ switches (Here, 4 switches per stage)

Key Characteristics

- Multistage interconnection network
- $N = 2^n$ processors
- $\log_2 N + 1$ stages
- Each stage has $N/2$ switching elements
- Each switch is a 2 × 2 switching element
- Unique path between any input and output
- Deterministic routing based on destination address bits

How It Works

1. Data enters the network from input processors at Stage 0.
2. At each stage, the switch either keeps the data on the same path (straight link) or changes it to the cross path (cross link) based on routing.
3. After passing through all stages, the data reaches the desired output processor at the output stage.

Important Notes

- The butterfly network is a blocking network.
- The number of links grows as $n \times 2^n$.
- Each connection between adjacent stages carries one data item per time slot.
- Widely used in parallel computers, FFT computations, and switching systems.

Applications

- Parallel processing systems
- Fast Fourier Transform (FFT)
- Switching and routing fabrics
- Interconnection networks in multiprocessor architectures
- Signal processing systems

2. Need for Butterfly Networks

As the number of processors in parallel systems increases, efficient and scalable communication becomes essential. Traditional interconnection schemes often lead to excessive hardware cost, long communication delays and poor scalability. Butterfly networks provide a cost–effective and high–performance solution.

1 Excessive Links in Fully Connected Networks

- A fully connected network requires $N(N-1)$ links for N processors, which becomes impractical as N grows. Hardware cost and wiring complexity increase rapidly.

2 High Communication Delay

- Long paths and multiple hops in traditional networks increase communication latency and reduce overall system performance.

3 Poor Scalability

- Many conventional topologies (bus, ring, tree) do not scale well beyond a certain size due to increased congestion and limited bandwidth.

4 High Hardware Cost

- More processors lead to more wires, connectors and switches in traditional networks, resulting in higher hardware cost and power consumption.

5 Bottleneck and Congestion

- Shared links in traditional networks become bottlenecks when many processors communicate simultaneously.

6 Limited Routing Flexibility

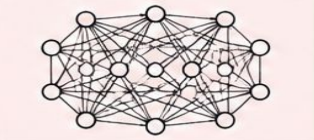
- Some topologies offer limited routing options, reducing fault tolerance and adaptability to dynamic traffic.

7 Need for Fast and Efficient Communication

- Parallel algorithms (e.g., FFT, scientific simulations) require fast, deterministic communication with low latency and predictable performance.

Figure 6.2 – Why Butterfly Network?

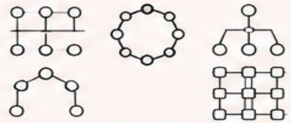
Fully Connected Network



- Requires $N(N-1)$ links
- Very high cost
- Complex wiring
- Hard to implement

→ Not Practical for Large N

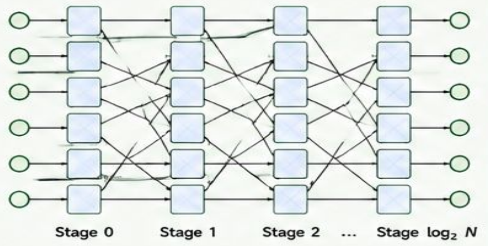
Traditional Topologies (Bus, Ring, Tree, Mesh)



- Longer communication paths
- Limited bandwidth
- Bottlenecks and congestion
- Poor scalability
- Higher latency

→ Not Efficient for Large Parallel Systems

Butterfly Network



- Requires only $N \log_2 N$ links
- Low hardware cost
- Logarithmic number of stages
- Deterministic routing
- Low latency
- Highly scalable
- Ideal for parallel systems
- Used in FFT and high–speed systems

Conclusion

Butterfly networks provide an optimal trade–off between cost and performance. They achieve fast, scalable and deterministic communication with a small number of switches and links, making them ideal for large parallel systems.

3. Organization of Butterfly Network

A butterfly network is a multistage interconnection network (MIN) that connects $N = 2^n$ processors through $n + 1$ stages of switching elements. Each stage transforms the information by allowing data to either go straight or cross to the next stage. The network is symmetric about its center.

Key Structural Characteristics

- Number of processors : $N = 2^n$
- Number of stages : $n + 1 = \log_2 N + 1$
- Number of switch elements : $(N/2) \log_2 N$
- Each stage contains $N/2$ switching elements (each is 2×2).
- Each processor has one incoming link and one outgoing link (except at the first and last stages where they are external).
- Data between any source and destination travels through exactly one unique path.

Example: $N = 8$ ($n = 3$)

Processors	: 8
Stages	: 4 (0, 1, 2, 3)
Switches / stage	: 4
Total switches	: $(8/2) \log_2 8 = 12$
Links	: $n \times N = 24$

General Formulas

Parameter	Formula
Number of processors	$N = 2^n$
Number of stages	$n + 1 = \log_2 N + 1$
Switches per stage	$N/2$
Total number of switches	$(N/2) \log_2 N$
Total number of links	$n \times N$
Diameter (maximum hops)	$n = \log_2 N$
Node degree (internal)	4 (two in, two out)

Stages in Butterfly Network

- Stage 0 : Input stage (sources enter)
- Stages $1 \ldots (n - 1)$: Middle stages
- Stage n : Output stage (destinations exit)
- There are n horizontal levels of processors.
- Each level has N nodes.
- Every processor in a level connects to exactly one processor in the next level.

Important Observations

1. The network is self-similar and recursive in structure.
2. The number of switches grows as $(N/2) \log_2 N$, much less than fully connected network $N(N - 1)/2$.
3. Communication cost (in hops) between any source and destination is $\log_2 N$.
4. The structure is symmetric with respect to the central stages.

How It Scales

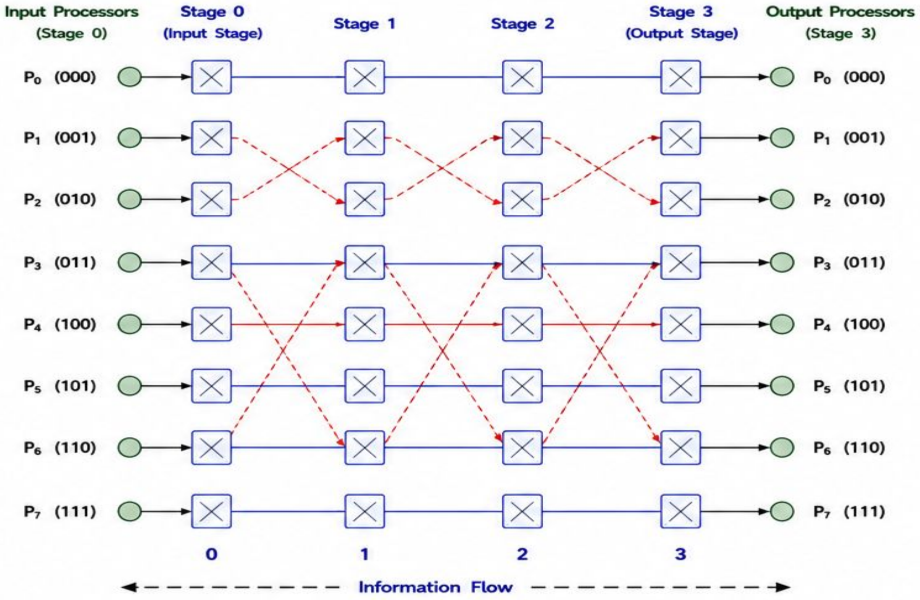
N (Processors)	n	Stages ($n + 1$)	Switches per Stage ($N/2$)	Total Switches $(N/2) \log_2 N$	Links ($n \times N$)
4	2	3	2	4	8
8	3	4	4	12	24
16	4	5	8	32	64
32	5	6	16	80	160

Legend

- Processor (Input / Output)
- Switching Element (2×2)
- Straight Link (No change in bit)
- Cross Link (Bit change)

Figure 6.3 – Organization of Butterfly Network

Example: Butterfly Network with $N = 8$ ($n = 3$)
The network has $n + 1 = 4$ stages (0 to 3) and $(N/2) \log_2 N = 12$ switches.



Structure Summary ($N = 8$, $n = 3$)

- 8 input processors enter at Stage 0.
- Data passes through 4 stages ($0 \rightarrow 1 \rightarrow 2 \rightarrow 3$).
- At each stage, every pair of processors exchanges data according to the routing decision.
- Each switch is a 2×2 switching element with straight or cross connections.
- Data reaches the desired output processor at Stage 3.

4. Components of Butterfly Network

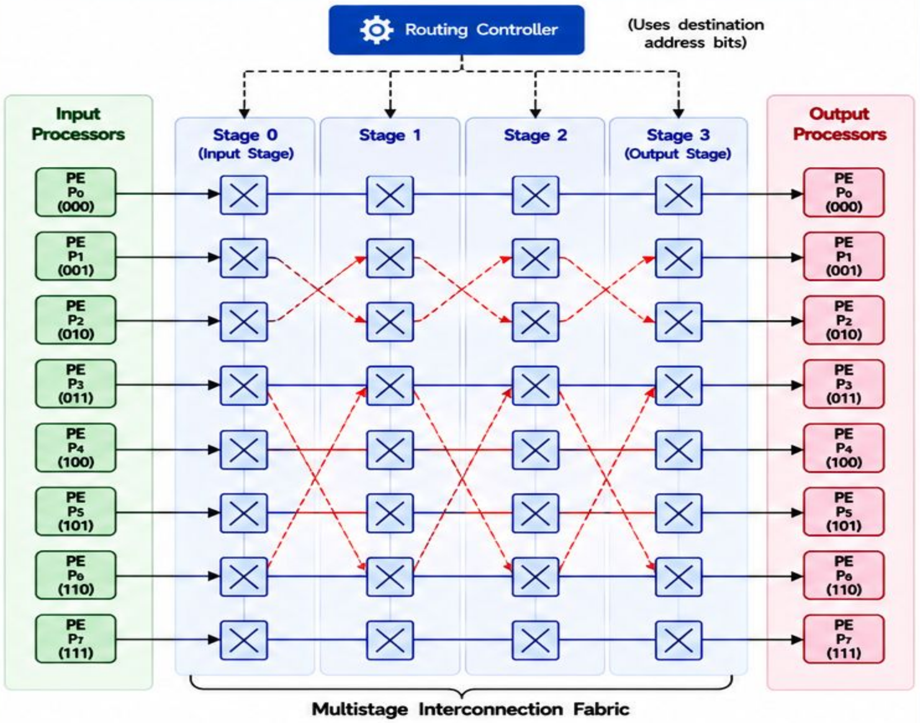
A butterfly network is built using a set of simple yet powerful components that work together to provide fast, scalable and deterministic communication among processors.

Component	Symbol / Icon	Description
1 Processing Element (PE)	PE	• Represents a processor/sending/receiving node. • Generates and consumes data. • Connected to the first and last stages of the network.
2 Switching Element (SE)		• Basic building block of the network. • A 2 × 2 switch with two inputs and two outputs. • Forwards data either straight or cross based on routing decision.
3 Communication Links		• Connect processors and switching elements between adjacent stages. • Straight link : transfers data without bit change. • Cross link : transfers data with bit change.
4 Stages	Stage *i*	• Arranged in $n + 1$ stages for $N = 2^n$ processors. • Each stage contains $N/2$ switching elements. • Transforms information one bit at a time.
5 Input Nodes		• Entry points of data into the network. • Located at Stage 0 (left side). • Each input node has a unique binary address.
6 Output Nodes		• Exit points of data from the network. • Located at the last stage (right side). • Receive data from the switching elements.
7 Routing Controller		• Determines the direction (straight or cross) of data at each switching element. • Uses the destination address bits for routing. • Ensures unique path from source to destination.
8 Interconnection Fabric		• Collection of all stages, switches and links. • Provides multistage connectivity between input and output processors. • Achieves logarithmic communication complexity.

Key Takeaways

- Butterfly network = Processors + Switching elements + Links + Stages + Routing.
- Each component plays a specific role in achieving fast and scalable communication.
- The simple 2 × 2 switch is the fundamental building block of the entire network.

Figure 6.4 – Components of Butterfly Network



Legend

- PE — Processing Element (Input / Output Processor)
- Switching Element (2 × 2)
- Straight Link (No change in bit)
- Cross Link (Bit change)
- Control Signal (from Routing Controller)

How the Components Work Together

1. Input processors generate data with destination address.
2. Routing controller checks the destination address bits.
3. Each switching element forwards data through straight or cross link based on routing decision.
4. Data moves stage by stage, modifying one bit per stage.
5. Data reaches the desired output processor through a unique path at the final stage.

5. Node Addressing Scheme

In a butterfly network with $N = 2^n$ processors, each processor is assigned an n-bit binary address. Routing from a source to a destination is performed by examining one bit of the destination address at each stage (from MSB to LSB). The bit examined at stage i $(0 \le i < n)$ decides whether the data takes the straight path (bit = 0) or the cross path (bit = 1) through that stage.

1 Addressing Rules

- $N = 2^n$ processors → n-bit addresses (0 to $N - 1$)
- Stages are numbered from 0 (input stage) to n (output stage)
- At stage i, the i^{th} bit (from MSB) of the destination address is used for routing.
- 0 → Straight (no change in the current bit)
- 1 → Cross (complement the current bit)
- One bit is resolved at each stage.

2 Node Notation

A node in a butterfly network is represented as:

(Stage , Address)

where Stage ∈ {0, 1, ..., n}
Address is an n-bit binary number.

3 Example: $N = 8$ $(n = 3)$

- Address length = 3 bits
- Stages = n + 1 = 4 (0, 1, 2, 3)
- Stage 0 : examines bit 2 (MSB)
- Stage 1 : examines bit 1
- Stage 2 : examines bit 0 (LSB)
- Stage 3 : Output stage (destination reached)

4 Node Address Table $(N = 8)$

Decimal	Binary Address	Decimal	Binary Address
0	000	4	100
1	001	5	101
2	010	6	110
3	011	7	111

5 Key Points

- Each stage resolves one bit of the destination address.
- After n stages, all bits are resolved and the data reaches the destination.
- The addressing scheme ensures a unique path between any source and destination.
- The scheme is simple, systematic and hardware-efficient.

Routing Decision at Stage i

Let the destination address be $d_{n-1}d_{n-2} \ldots d_1d_0$ (indexed from MSB to LSB).

At stage i $(0 \le i < n)$:

Bit d_{n-1-i}	Action	Path
0	Keep current bit	Straight
1	Complement current bit	Cross

The current node address is updated by considering the bit d_{n-1-i}.

After n stages, the packet arrives at the node whose address equals the destination address.

Example Summary $(N = 8)$

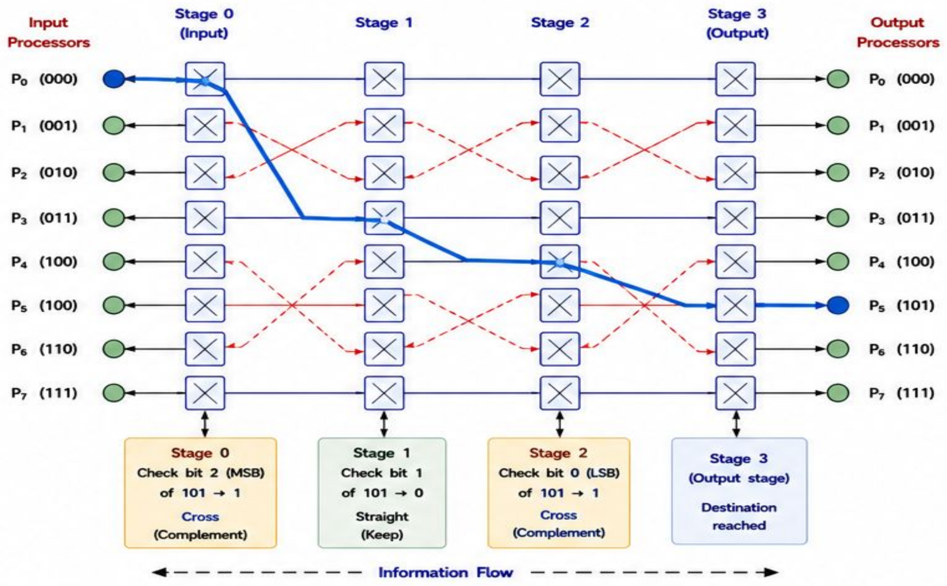
- n = 3 bits → addresses 000 to 111
- Stages : 0, 1, 2, 3
- At each stage, one bit is checked from MSB → LSB
- After 3 decisions, destination is reached.

Benefits

✓ Simple and regular addressing
✓ Easy hardware implementation
✓ Deterministic routing
✓ Ensures one unique path

Figure 6.5 – Node Addressing Scheme

Example: Routing from Source (000) to Destination (101) in an 8-node Butterfly Network $(n = 3)$
At each stage, the corresponding bit of the destination address (101) is examined.



Routing Table for Source (000) → Destination (101)

Stage (i)	Current Node Address	Destination Address (101)	Bit Examined (d_{2-i})	Bit Value	Decision	Next Node Address	Operation
0	000	101	d_2 (MSB)	1	Cross (Complement)	100	1^{st} bit resolved
1	100	101	d_1	0	Straight (Keep)	100	2^{nd} bit resolved
2	100	101	d_0 (LSB)	1	Cross (Complement)	101	3^{rd} bit resolved
3	101	101	–	–	–	101	Destination reached

Observations: One bit is resolved at each stage (MSB → LSB). After n = 3 stages, the packet reaches the destination node 101.

6. Communication in Butterfly Networks

A butterfly network is a multistage interconnection network where data is transferred from a source processor to a destination processor through a sequence of switching stages. At every stage, the packet is forwarded either through a straight link (no change in the current bit being examined) or a cross link (complement of the current bit). Thus, communication proceeds stage by stage until the packet reaches the destination.

Key Characteristics of Communication

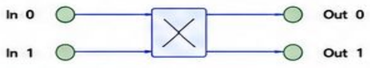
- Communication is unidirectional across stages (from Stage 0 to Stage n).
- At each stage, one bit of the destination address is examined (MSB → LSB).
- Based on the bit value, the packet takes a straight or cross path.
- A unique path exists between any source and destination.
- Communication complexity is $O(\log N)$.

Example Network

- $N = 8$ processors ($n = 3$)
- Stages $= n + 1 = 4$ (0, 1, 2, 3)
- Switches per stage $= N/2 = 4$
- Communication links $= n \times N = 24$

Types of Communication at a Stage

1. Straight Communication



- Occurs when the examined bit = 0.
- The packet continues through the same upper or lower output.
- No change in the current bit.

2. Cross Communication

- Occurs when the examined bit = 1.
- The packet takes the opposite (output) path.
- Bit is complemented (0 ↔ 1).

Communication Process (Stage by Stage)

1. Source processor generates a packet with the destination address.
2. At Stage i, the routing controller examines the i^{th} bit (from MSB to LSB) of the destination.
3. If bit = 0 → packet takes straight link.
4. If bit = 1 → packet takes cross link.
5. After n stages, the packet reaches the destination processor.

Information Flow

Stage 0 → Stage 1 → Stage 2 → … → Stage n (Output Stage)

Illustrative Example ($N = 8$)

Stage	Bit Examined	Bit Value (Example: 101)	Communication Type
Stage 0	MSB (d_2)	1	Cross
Stage 1	Next bit (d_1)	0	Straight
Stage 2	Next bit (d_0)	1	Cross
Stage 3	–	–	Destination reached

Important Note

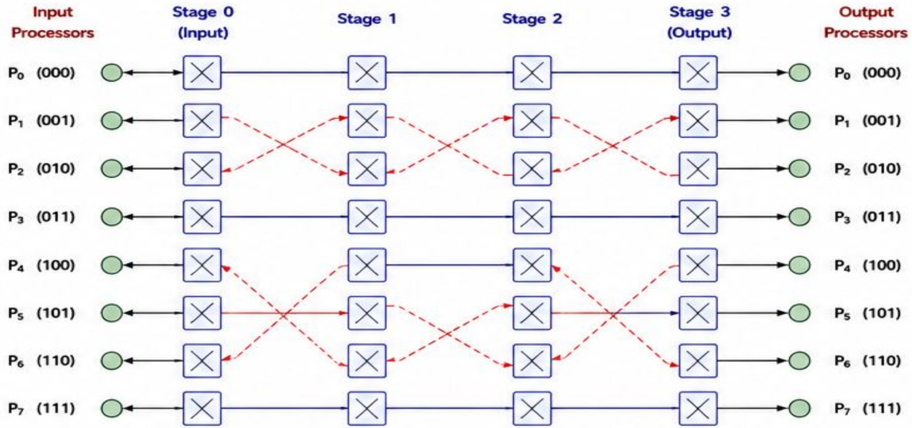
At each stage exactly one bit is resolved. Hence after n stages, all bits are resolved and the packet reaches the required destination.

Key Takeaway

Butterfly networks achieve fast and efficient communication by resolving one address bit at each stage, resulting in a unique path and low communication complexity of $O(\log N)$.

Figure 6.6 – Communication Paths

Example: Communication paths in an 8-node Butterfly Network ($n = 3$, 4 stages)



Straight Link (bit = 0) Cross Link (bit = 1) 2 × 2 Switching Element

Example Communication Paths ($N = 8$)

Path 1: 000 → 000 (Straight, Straight, Straight)

Bit sequence: 1 0 0

All bits = 0 → all straight links

Path 2: 000 → 101 (Cross, Straight, Cross)

Bit sequence: 1 0 1

Cross → Straight → Cross

Path 3: 011 → 110 (Straight, Cross, Cross)

Bit sequence: 1 1 0

Straight → Cross → Cross

Summary of Communication

Aspect	Description
Direction	Always from left (input stage) to right (output stage)
Bit Resolution	One destination address bit is resolved at each stage (MSB → LSB)
Path Uniqueness	Exactly one unique path exists between any input and output processor
Communication Cost	Number of hops = number of stages = $n = \log_2 N$
Throughput	One data item per link per time slot (assuming synchronous operation)

Key Points

- ✓ Straight link → bit = 0
- ✓ Cross link → bit = 1
- ✓ n stages resolve n bits
- ✓ Unique path
- ✓ Communication cost = $O(\log N)$

7. Routing Algorithm

Butterfly networks use deterministic, bit-wise routing. At each stage, the routing controller examines one bit of the destination address (from MSB to LSB).

- If the bit = 0 → take STRAIGHT link (no change in current bit).
- If the bit = 1 → take CROSS link (complement the current bit).

Routing Algorithm (n-bit Butterfly Network)

1. Read the n-bit destination address $D = d_{n-1}d_{n-2} \cdots d_1d_0$.
2. Start at Stage 0 with the source processor.
3. For stage $i = 0$ to $n - 1$
   - Examine bit d_{n-1-i} of the destination address. (i.e., bits are examined from MSB to LSB).
   - If $d_{n-1-i} = 0$ → select STRAIGHT link.
   - If $d_{n-1-i} = 1$ → select CROSS link.
   - Move to the next stage.
4. After n stages, the packet reaches the destination.
5. Exactly one bit is resolved at each stage.
6. The path taken is unique for each source–destination pair.

Routing Decision Rule

Let the destination address bits be

$D = d_{n-1}d_{n-2} \cdots d_1d_0$ (MSB → LSB)

At stage i,

- If $d_{n-1-i} = 0$ → STRAIGHT (Keep bit)
- If $d_{n-1-i} = 1$ → CROSS (Complement bit)

Straight (bit = 0)
The current bit remains unchanged. Packet continues on the same row in the next stage.

Cross (bit = 1)
The current bit is complemented (0 ↔ 1). Packet moves to the opposite row in the next stage.

Example Walkthrough ($N = 8$, $n = 3$)

Source = 000 Destination = 101 ($d_2d_1d_0$ = 1 1)

We examine bits from MSB to LSB: 1, 0, 1

Stage (i)	Bit Examined (d_{2-i})	Bit Value	Decision	Operation	Next Address	Path Type
0 (Input)	d_2 (MSB)	1	CROSS	Complement bit 2	100	Cross
1	d_1	0	STRAIGHT	Keep bit 1	100	Straight
2	d_0 (LSB)	1	CROSS	Complement bit 0	101	Cross
3 (Output)	–	–	–	Destination reached	101	–

General Properties

- ✓ One bit is decided at each stage.
- ✓ Total number of stages = n.
- ✓ Routing complexity = $O(n) = O(\log_2 N)$.
- ✓ Path between any source and destination is unique.
- ✓ The number of cross links used depends on the Hamming distance between source and destination.

Routing Pseudocode

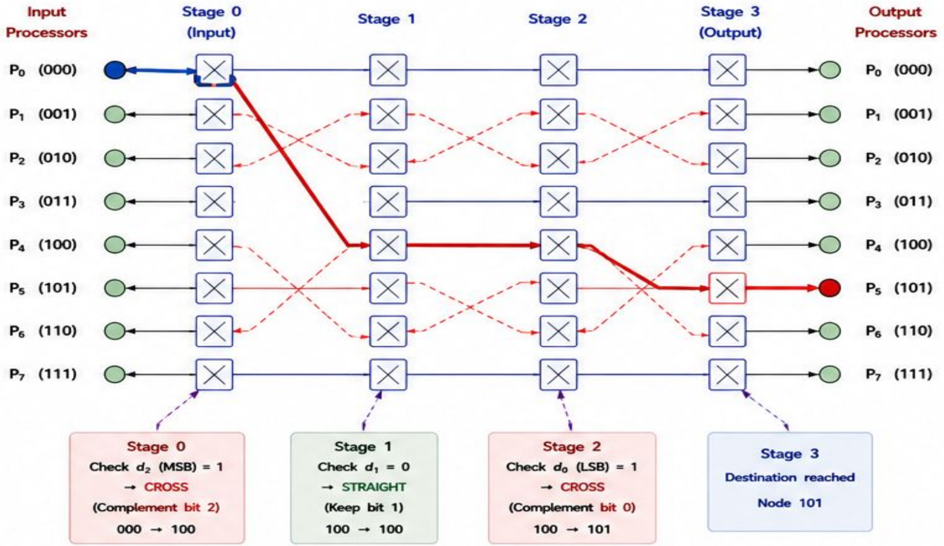
```
Input: Source S, Destination D (n-bit)
current ← S
for i = 0 to n − 1 do
    if bit_{n-1-i}(D) = 0 then
        // Straight
        current_{n-1-i} ← current_{n-1-i}
    else
        // Cross
        current_{n-1-i} ← 1 − current_{n-1-i}
Output: current (Destination)
```

Key Takeaway

Butterfly routing is simple, deterministic, and efficient:
At each stage, decide the next hop by examining one bit of the destination address.

Figure 6.7 – Routing in Butterfly Network

Example: Route a packet from Source 000 to Destination 101 in an 8-node Butterfly Network ($n = 3$)
Destination address = 101 (bits examined from MSB to LSB: 1, 0, 1)



Straight Link (bit = 0) | Cross Link (bit = 1) | 2 × 2 Switching Element
Source (000) | Destination (101) | Routing Decision

Routing Summary (000 → 101)

Stage (i)	Bit Examined	Bit Value	Decision	Operation	Next Address	Path Type
0	d_2 (MSB)	1	CROSS	Complement bit 2	100	Cross
1	d_1	0	STRAIGHT	Keep bit 1	100	Straight
2	d_0 (LSB)	1	CROSS	Complement bit 0	101	Cross
3	–	–	–	Destination reached	101	–

Observations

- One bit is resolved at each stage.
- After $n = 3$ stages, all bits are resolved.
- Unique path from 000 to 101.
- Number of cross links used = number of 1's in destination address (here = 2).

☆ Routing Complexity: $O(n) = O(\log_2 N)$ | Number of Stages: n | Path Uniqueness: Yes | Blocking: Yes

8. Performance Analysis

The performance of a butterfly network depends on the number of stages, switching elements, links, diameter, node degree and communication complexity. For a butterfly network with $N = 2^n$ processors, the major performance parameters are derived as below.

	Parameter	Description	Formula ($N = 2^n$)	Value for $N = 8$ ($n = 3$)	Remarks
1	Number of Processors (N)	Total number of processing elements connected.	$N = 2^n$	8	n = number of address bits.
2	Number of Stages (S)	Number of switching stages between input and output.	$S = n + 1$ (0 to n)	4	Includes input stage (0) and output stage (n).
3	Number of Switching Elements (SE)	Total number of 2×2 switching elements.	$SE = \frac{N}{2} \log_2 N = n \cdot 2^{n-1}$	$\frac{8}{2} \log_2 8 = 12$ ($3 \times 4 = 12$)	Each stage has $N/2$ switches.
4	Number of Communication Links (L)	Total number of links (Straight + Cross).	$L = n \cdot N = N \log_2 N$	$3 \times 8 = 24$	Each stage contributes N links.
5	Diameter (D)	Maximum number of hops between any two nodes.	$D = n = \log_2 N$	3	Longest path length.
6	Node Degree	Number of links connected to a node (bidirectional).	Typically 4 (2 inputs + 2 outputs)	4	Interior nodes have degree 4.
7	Communication Complexity	Number of hops required for message transfer.	$O(\log_2 N)$	$O(\log_2 8) = O(3)$	Logarithmic in N.
8	Bisection Bandwidth (B_w)	Maximum number of links that can be cut to divide the network into two halves.	$B_w = N$	8	Excellent bandwidth scalability.
9	Hardware Cost	Proportional to number of switches and links.	$O(N \log_2 N)$	$O(8 \log_2 8) = O(24)$	Much cheaper than fully connected.
10	Scalability	Ability to support more processors.	Excellent	–	Adding one more bit doubles N.

Key Insights

- Number of stages, diameter and communication complexity grow logarithmically with N.
- Hardware cost grows as $O(N \log N)$, far lower than the fully connected network $O(N^2)$.
- Bisection bandwidth grows linearly with N, providing good throughput.
- Each node has fixed degree (4), simplifying switch design.

Example ($N = 8$, $n = 3$)

- Stages = $n + 1 = 4$
- Switches = 12
- Links = 24
- Diameter = 3
- Node Degree = 4
- Communication Complexity = $O(3)$

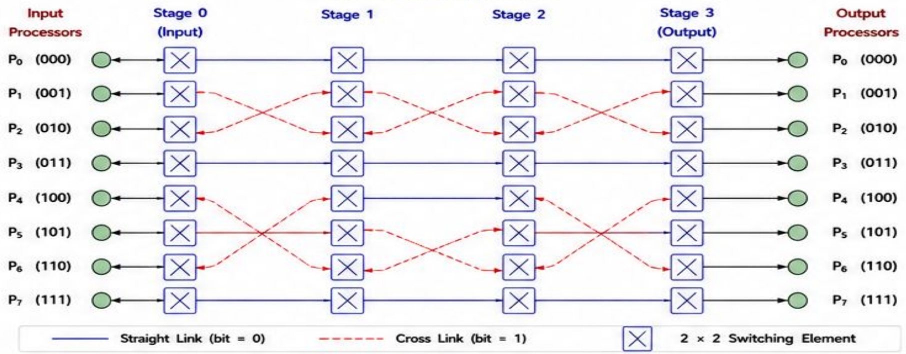
Summary

Butterfly networks offer logarithmic routing cost, high scalability and efficient hardware utilization, making them suitable for large-scale parallel systems and FFT-based computations.

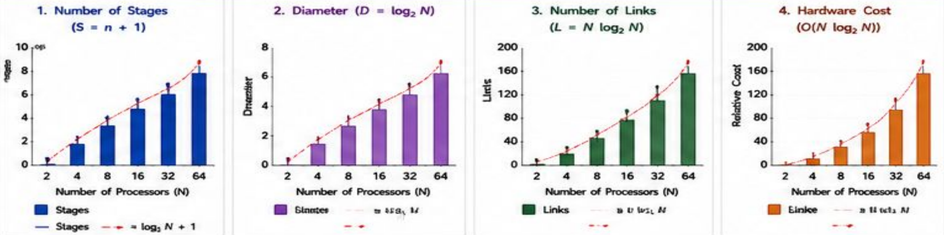
Figure 6.8 – Performance Metrics

Example: 8-Node Butterfly Network ($N = 8$, $n = 3$)
Stages = 4 (0 to 3), Switches = 12, Links = 24

8-Node Butterfly Network



Performance Metric Comparison



Other Key Parameters (for $N = 8$)

Bisection Bandwidth (B_w)	8 links
Node Degree	4 (2 in + 2 out)
Switching Elements	12
Communication Complexity	$O(\log_2 N) = O(3)$
Path Uniqueness	Unique path
Network Type	Blocking
Scalability	Excellent

Performance Highlights

- ✓ Logarithmic number of stages and diameter.
- ✓ Communication complexity = $O(\log_2 N)$.
- ✓ Linear bisection bandwidth ensures good throughput.
- ✓ Hardware cost grows as $O(N \log_2 N)$, much lower than fully connected network $O(N^2)$.
- ✓ Deterministic routing with a unique path between any source and destination.

Comparison with Other Topologies (for $N = 8$)

Topology	Diameter	Links	Node Degree	Blocking
Butterfly	3	24	4	Yes
Hypercube	3	32	3	No
Omega	3	24	4	Yes
Fully Connected	1	56	7	No

Key Takeaway

Butterfly networks achieve high performance with logarithmic routing cost and $O(N \log N)$ hardware complexity, providing an excellent balance between cost, speed and scalability for parallel systems.

9. Advantages

Butterfly networks provide an efficient and structured interconnection scheme that is well suited for parallel and distributed systems.

1	**Low Cost**	Requires only $\frac{N}{2}\log_2 N$ switching elements and $N\log_2 N$ links, much less than the fully connected network ($N(N-1)$ links).
2	**Scalability**	Performance grows logarithmically with N. Suitable for large-scale systems and FFT computations.
3	**Deterministic Routing**	Simple bit-based routing provides a unique path between any source–destination pair. Easy to implement in hardware.
4	**Regular Structure**	Highly regular and symmetric topology simplifies design, layout and verification.
5	**High Throughput**	Multiple communications can occur simultaneously on different paths with minimal contention.
6	**Efficient Hardware Utilization**	Each switch is a simple 2×2 element with low hardware complexity and fast operation.
7	**Good for Parallel Algorithms**	Ideal for FFT, sorting, prefix computations and other data permutation intensive algorithms.

10. Disadvantages

Despite many benefits, butterfly networks also have certain limitations that need to be considered.

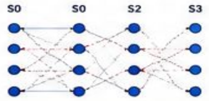
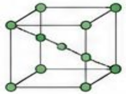
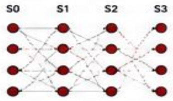
1	**Blocking**	Multiple packets may compete for the same link, causing blocking. Not a rearrangeable (Clos) network.
2	**Limited Path Diversity**	Only one unique path exists between a source and destination. No alternative path in case of link failure.
3	**Not Fault Tolerant**	Failure of a link or switch can disconnect communication without any alternate route.
4	**Increased Latency with Size**	Diameter is $n = \log_2 N$; latency increases logarithmically as network size grows.
5	**Traffic Patterns Matter**	Performance degrades for certain traffic patterns due to higher contention on some links.
6	**Not Reconfigurable**	Fixed topology cannot adapt to dynamic communication requirements.

11. Applications

Butterfly networks are widely used in systems where large amounts of data need to be permuted or communicated efficiently.

1. **FFT Computations** — Efficient data permutation in radix-2 FFT algorithms.
2. **Sorting and Ranking** — Used in sorting networks and parallel ranking algorithms.
3. **Matrix Transposition** — Ideal for transposing matrices in parallel computing systems.
4. **Parallel Processing Systems** — Interconnection backbone for multiprocessors and many-core architectures.
5. **Digital Signal Processing** — Used in convolution, correlation and filter bank implementations.
6. **Data Rearrangement** — General purpose data permutation and reordering.
7. **Multimedia and Image Processing** — Pixel reordering in transforms such as DCT, wavelets, etc.
8. **Cryptography** — Used in permutation and scrambling operations.
9. **Hardware Accelerators** — Interconnect fabric in ASIC/FPGA-based accelerators.
10. **Educational & Research** — Benchmark topology for studying interconnect performance and algorithms.

12. Comparison with Hypercube and Omega Networks

Feature	Butterfly Network ($N = 2^n$)	Hypercube Network ($N = 2^n$)	Omega Network ($N = 2^n$)
Topology ($N = 8$ example)			
Number of Stages	$n = \log_2 N$	$n = \log_2 N$	$n = \log_2 N$
Switching Elements	$\frac{N}{2}\log_2 N$	$\frac{N}{2}\log_2 N$	$\frac{N}{2}\log_2 N$
Number of Links	$N\log_2 N$	$\frac{N}{2}\log_2 N$	$N\log_2 N$
Node Degree	4 (bidirectional)	$n = \log_2 N$	4 (bidirectional)
Diameter	$n = \log_2 N$	$n = \log_2 N$	$2n - 1 = 2\log_2 N - 1$
Path Between Any Two Nodes	Unique path	Multiple paths ($n!$ / 2 shortest paths)	Unique path
Blocking	Yes	No	No (rearrangeable)
Fault Tolerance	Low	High	High
Hardware Cost	Low	Moderate	Moderate
Best Suited For	FFT, permutations, data rearrangement	General purpose communication	High performance, rearrangeable traffic

Key Takeaway

- Butterfly networks are cost-effective with simple structure and unique paths.
- Hypercubes offer multiple paths and high fault tolerance but have higher node degree.
- Omega networks are rearrangeable and non-blocking with slightly higher diameter.

Legend

Straight link (bit = 0)

Cross link (bit = 1)

13. Worked Example

Problem Statement

In an 8-node butterfly network ($N = 8$, $n = 3$), find the path when source processor P0 (000) sends data to destination processor P5 (101). Show the routing decision at each stage.

Step 1: Network Parameters

- $N = 2^n = 8$ processors → $n = 3$ bits
- Number of stages, $S = n + 1 = 4$ (0, 1, 2, 3)
- Switches per stage, $SE = N/2 = 4$
- Communication links, $L = N \log_2 N = 8 \times 3 = 24$

Step 2: Addressing

Processor	P0	P1	P2	P3	P4	P5	P6	P7
Address (3 bits)	000	001	010	011	100	101	110	111

Step 3: Routing Table (Source 000 → Destination 101)

Stage (i)	Bit Examined (d_{2-i})	Bit Value	Decision	Operation	Next Node Address	Path Type
0 (Input)	d_2 (MSB)	1	CROSS	Complement bit 2	100	Cross
1	d_1	0	STRAIGHT	Keep bit 1	100	Straight
2	d_0 (LSB)	1	CROSS	Complement bit 0	101	Cross
3 (Output)	—	—	—	Destination reached	101	—

Step 4: Path Summary

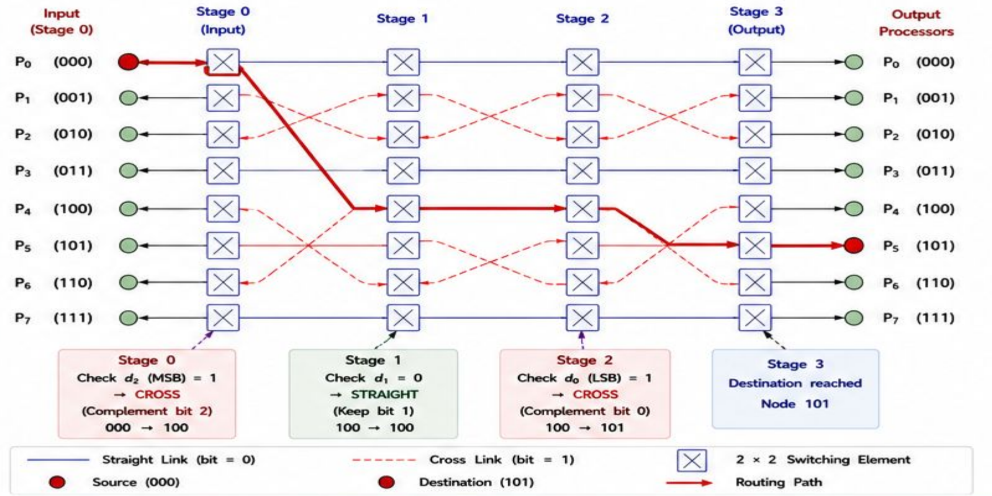
- Path: 000 → 100 → 100 → 101
- Link sequence: Cross, Straight, Cross
- Total hops = 3 (equals number of stages − 1)
- Unique path between source and destination.

Step 5: Observations

- ✓ One bit is resolved at each stage (MSB → LSB).
- ✓ After 3 stages, all bits match the destination 101.
- ✓ Deterministic and deadlock-free routing.

Figure 6.10 – Worked Example

Example: Route data from Source P0 (000) to Destination P5 (101) in an 8-node Butterfly Network



14. Real-World Case Study

Butterfly networks are widely used in interconnection networks for parallel computers and high-performance systems.

Application: High-Performance Computing (HPC) Systems

Many parallel computers use butterfly networks as the communication backbone to connect processors efficiently.

Example Systems

- Intel Paragon, IBM SP2, and SGI Origin series (early systems)
- Modern HPC clusters and supercomputers use variants for scalable interconnects.

Why Butterfly Network?

- Provides low-cost, scalable, and deterministic routing.
- Suitable for FFT, matrix operations, and scientific simulations.
- Supports high throughput with simple switch hardware.

Impact

Enables efficient communication among thousands of processors, reducing computation time in large-scale scientific and engineering applications.

Typical Use Cases

- Scientific simulations (weather, fluid dynamics)
- Signal and image processing
- Large-scale data analytics
- AI/ML training on parallel architectures

15. Summary

Butterfly networks provide an efficient and structured communication scheme with logarithmic complexity and unique paths, making them ideal for parallel and distributed systems.

Key Highlights

Structure	Multistage interconnection network with $n + 1$ stages for $N = 2^n$ processors.
Routing	Deterministic bit-wise routing (MSB → LSB) using straight (0) or cross (1) links.
Performance	Number of stages = $n + 1$, switches = $\frac{N}{2}$ per stage, links = $N \log_2 N$, diameter = n.
Communication Complexity	$O(\log_2 N)$, logarithmic in number of processors.
Hardware Cost	$O(N \log_2 N)$, much lower than fully connected network.
Path Property	Provides a unique path between any source and destination.
Scalability	Highly scalable and suitable for large-scale systems.

Key Takeaways

- ✓ Simple and regular topology
- ✓ Low cost and high scalability
- ✓ Deterministic and deadlock-free routing
- ✓ One bit resolved at each stage
- ✓ Unique communication path
- ✓ Suitable for FFT, data permutations, and parallel computations
- ✓ Widely used in HPC and interconnection networks

> “Butterfly networks achieve the right balance between performance, cost, and scalability through structured multistage communication.”